

Obfuscated Files or Information

Sub-techniques (18)



Adversaries may attempt to make an executable or file difficult to discover or analyze by encrypting, encoding, or otherwise obfuscating its contents on the system or in transit. This is common behavior that can be used across different platforms and the network to evade defenses.

Payloads may be compressed, archived, or encrypted in order to avoid detection. These payloads may be used during Initial Access or later to mitigate detection. Sometimes a user's action may be required to open and [Deobfuscate/Decode Files or Information](#) for [User Execution](#). The user may also be required to input a password to open a password protected compressed/encrypted file that was provided by the adversary.^[1] Adversaries may also use compressed or archived scripts, such as JavaScript.

Portions of files can also be encoded to hide the plain-text strings that would otherwise help defenders with discovery.^[2] Payloads may also be split into separate, seemingly benign files that only reveal malicious functionality when reassembled.^[3]

Adversaries may also abuse [Command Obfuscation](#) to obscure commands executed from payloads or directly via [Command and Scripting Interpreter](#). Environment variables, aliases, characters, and other platform/language specific semantics can be used to evade signature based detections and application control mechanisms.^{[4][5][6]}

ID: T1027

Sub-techniques: [T1027.001](#), [T1027.002](#), [T1027.003](#), [T1027.004](#), [T1027.005](#), [T1027.006](#), [T1027.007](#), [T1027.008](#), [T1027.009](#), [T1027.010](#), [T1027.011](#), [T1027.012](#), [T1027.013](#), [T1027.014](#), [T1027.015](#), [T1027.016](#), [T1027.017](#), [T1027.018](#)

- ① **Tactic:** [Stealth](#)
- ① **Platforms:** ESXi, Linux, Network Devices, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0025	2016 Ukraine Electric Power Attack	During the 2016 Ukraine Electric Power Attack , Sandworm Team used heavily obfuscated code with Industroyer in its Windows Notepad backdoor. ^[7]
C0057	3CX Supply Chain Attack	During the 3CX Supply Chain Attack , AppleJeuS payloads use AES-256 GCM cipher to encrypt data to include ICONICSTEALER and VEILED SIGNAL. ^{[8][9]}
S1028	Action RAT	Action RAT 's commands, strings, and domains can be Base64 encoded within the payload. ^[10]
S0045	ADVSTORESHELL	Most of the strings in ADVSTORESHELL are encrypted with an XOR-based algorithm; some strings are also encrypted with 3DES and reversed. API function names are also reversed, presumably to avoid detection in memory. ^{[11][12]}
S0331	Agent Tesla	Agent Tesla has had its code obfuscated in an apparent attempt to make analysis difficult. ^[13] Agent Tesla has used the Rijndael symmetric encryption algorithm to encrypt strings. ^[14]
S1025	Amadey	Amadey has obfuscated strings such as antivirus vendor names, domains, files, and others. ^[15]
S0504	Anchor	Anchor has obfuscated code with stack strings and string encryption. ^[16]
S9027	ANELLDR	ANELLDR code implements anti-analysis techniques including control flow flattening and Mixed Boolean Arithmetic (MBA). ^[17]
S0584	AppleJeuS	AppleJeuS has XOR-encrypted collected system information prior to sending to a C2. AppleJeuS has also used the open source ADVObfuscation library for its components. ^[18]
S0622	AppleSeed	AppleSeed has the ability to Base64 encode its payload and custom encrypt API calls. ^[19]
G0099	APT-C-36	APT-C-36 has used ConfuserEx to obfuscate its variant of Imminent Monitor , compressed payloads and RAT packages, and password protected encrypted email attachments to avoid detection. ^[20] APT-C-36 has also compressed initial droppers into ZIP, LHA and UUE formats. ^[21]
G0022	APT3	APT3 obfuscates files or information to help evade defensive measures. ^[22]
G0067	APT37	APT37 obfuscates strings and payloads. ^{[23][24][25]}
G0096	APT41	APT41 used VMProtected binaries in multiple intrusions. ^[26]
S0640	Avaddon	Avaddon has used encrypted strings. ^[27]
S1053	AvosLocker	AvosLocker has used XOR-encoded strings. ^[28]
G0135	BackdoorDiplomacy	BackdoorDiplomacy has obfuscated tools and malware it uses with VMProtect. ^[29]
G0063	BlackOasis	BlackOasis 's first stage shellcode contains a NOP sled with alternative instructions that was likely designed to bypass antivirus tools. ^[30]
S1226	BOOKWORM	BOOKWORM has been delivered using self-extracting RAR archives. ^[31]
S0635	BoomBox	BoomBox can encrypt data using AES prior to exfiltration. ^[32]
S0651	BoxCaon	BoxCaon used the "StackStrings" obfuscation technique to hide malicious functionalities. ^[33]

ID	Name	Description
S1039	Bumblebee	Bumblebee has been delivered as password-protected zipped ISO files and used control-flow-flattening to obfuscate the flow of functions. ^{[39][40][41]}
S0482	Bundlore	Bundlore has obfuscated data with base64, AES, RC4, and bz2. ^[42]
S1118	BUSHWALK	BUSHWALK can encrypt the resulting data generated from C2 commands with RC4. ^[43]
C0015	C0015	During C0015 , the threat actors used Base64-encoded strings. ^[44]
C0017	C0017	During C0017 , APT41 broke malicious binaries, including DEADEYE and KEYPLUG , into multiple sections on disk to evade detection. ^[45]
S0030	Carbanak	Carbanak encrypts strings to make analysis more difficult. ^[46]
S0335	Carbon	Carbon encrypts configuration files and tasks for the malware to complete using CAST-128 algorithm. ^{[47][48]}
S0465	CARROTBALL	CARROTBALL has used a custom base64 alphabet to decode files. ^[49]
S1149	CHIMNEYSWEEP	CHIMNEYSWEEP can use a custom Base64 alphabet to encode an API decryption key. ^[50]
S0660	Clambling	The Clambling executable has been obfuscated when dropped on a compromised host. ^[51]
S1105	COATHANGER	COATHANGER can store obfuscated configuration information in the last 56 bytes of the file <code>/date/.bd.key/preload.so</code> . ^[52]
S0154	Cobalt Strike	Cobalt Strike can hash functions to obfuscate calls to the Windows API and use a public/private key pair to encrypt Beacon session metadata. ^{[53][54]}
S0369	CoinTicker	CoinTicker initially downloads a hidden encoded file. ^[55]
S0244	Comnie	Comnie uses RC4 and Base64 to obfuscate strings. ^[56]
S0126	ComRAT	ComRAT has encrypted its virtual file system using AES-256 in XTS mode. ^{[57][58]}
S0608	Conficker	Conficker has obfuscated its code to prevent its removal from host machines. ^[59]
S0575	Conti	Conti can use compiler-based obfuscation for its code, encrypt DLLs, and hide Windows API calls. ^{[60][61][62]}
S0137	CORESHELL	CORESHELL obfuscates strings using a custom stream cipher. ^[63]
S0625	Cuba	Cuba has used multiple layers of obfuscation to avoid analysis, including its Base64 encoded payload. ^[64]
S1111	DarkGate	DarkGate uses a hard-coded string as a seed, along with the victim machine hardware identifier and input text, to generate a unique string used as an internal mutex value to evade static detection based on mutexes. ^[65]
S1066	DarkTortilla	DarkTortilla has been obfuscated with the DeepSea .NET and ConfuserEx code obfuscators. ^[66]
S0187	Daserf	Daserf uses encrypted Windows APIs and also encrypts data using the alternative base64+RC4 or the Caesar cipher. ^[67]
S0354	Denis	Denis obfuscates its code and encrypts the API names. ^[68]

ID	Name	Description
S0062	DustySky	The DustySky dropper uses a function to obfuscate the name of functions and other parts of the malware. ^[73]
G1006	Earth Lusca	Earth Lusca used Base64 to encode strings. ^[74]
S0377	Ebury	Ebury has obfuscated its strings with a simple XOR encryption with a static key. ^[75]
S0593	ECCENTRICBANDWAGON	ECCENTRICBANDWAGON has encrypted strings with RC4. ^[76]
S0624	Ecipekac	Ecipekac can use XOR, AES, and DES to encrypt loader shellcode. ^[77]
S0605	EKANS	EKANS uses encoded strings in its process kill list. ^[78]
S0091	Epic	Epic heavily obfuscates its code to make analysis more difficult. ^[79]
S0512	FatDuke	FatDuke can use base64 encoding, string stacking, and opaque predicates for obfuscation. ^[80]
S0355	Final1stspy	Final1stspy obfuscates strings with base64 encoding. ^[81]
S0182	FinFisher	FinFisher is heavily obfuscated in many ways, including through the use of spaghetti code in its functions in an effort to confuse disassembly programs. It also uses a custom XOR algorithm to obfuscate code. ^{[82][83]}
S0696	Flagpro	Flagpro has been delivered within ZIP or RAR password-protected archived files. ^[84]
S9033	Fooder	Fooder has stored its embedded payload in encrypted form within the binary, using a hardcoded key modified at runtime to produce the AES decryption key. ^[85]
G0093	GALLIUM	GALLIUM used a modified version of HTRAN in which they obfuscated strings such as debug messages in an apparent attempt to evade detection. ^[86]
G0084	Gallmaker	Gallmaker obfuscated shellcode used during execution. ^[87]
G0047	Gamaredon Group	Gamaredon Group has delivered self-extracting 7z archive files within malicious document attachments. ^[88] Additionally, Gamaredon Group has used an obfuscated .drv file. ^[89]
S1138	Gootloader	The Gootloader first stage script is obfuscated using random alpha numeric strings. ^{[90][91]}
S0690	Green Lambert	Green Lambert has encrypted strings. ^{[92][93]}
S0632	GrimAgent	GrimAgent has used Rotate on Right (RoR) and Rotate on Left (RoL) functionality to encrypt strings. ^[94]
S0132	H1N1	H1N1 uses multiple techniques to obfuscate strings, including XOR. ^[95]
S0499	Hancitor	Hancitor has used Base64 to encode malicious links. ^[96]
S0070	HTTPBrowser	HTTPBrowser 's code may be obfuscated through structured exception handling and return-oriented programming. ^[97]
S9007	HTTPTroy	HTTPTroy has obfuscated strings using Single Instruction Multiple Data (SIMD) instructions to hinder analysis and detection. ^[98]
S0203	Hydrag	Hydrag uses basic obfuscation in the form of spaghetti code. ^{[99][100]}

ID	Name	Description
S0260	InvisiMole	InvisiMole avoids analysis by encrypting all strings, internal files, configuration data and by using a custom executable format. ^{[102][103]}
S0189	ISMinjector	ISMinjector is obfuscated with the off-the-shelf SmartAssembly .NET obfuscator created by red-gate.com. ^[104]
S0201	JPIN	A JPIN uses a encrypted and compressed payload that is disguised as a bitmap within the resource section of the installer. ^[105]
S0283	jRAT	jRAT 's Java payload is encrypted with AES. ^[106] Additionally, backdoor files are encrypted using DES as a stream cipher. Later variants of jRAT also incorporated AV evasion methods such as Java bytecode obfuscation via the commercial Allatori obfuscation tool. ^[107]
S0265	Kazuar	Kazuar is obfuscated using the open source ConfuserEx protector. Kazuar also obfuscates the name of created files/folders/mutexes and encrypts debug messages written to log files using the Rijndael cipher. ^[108]
G0004	Ke3chang	Ke3chang has used Base64-encoded shellcode strings. ^[109]
S0607	KillDisk	KillDisk uses VMProtect to make reverse engineering the malware more difficult. ^[110]
G0094	Kimsuky	Kimsuky has obfuscated binary strings including the use of XOR encryption and Base64 encoding. ^[111] ^[112] Kimsuky has also modified the first byte of DLL implants targeting victims to prevent recognition of the executable file format. ^[113] Kimsuky has obfuscated strings using Single Instruction Multiple Data (SIMD) instructions that complicate static analysis. ^[98]
S0641	Kobalos	Kobalos encrypts all strings using RC4 and bundles all functionality into a single function call. ^[114]
S0681	Lizar	Lizar has obfuscated the fingerprint of the victim system, the local IP address, and the Fowler-Noll-V 1 (FNV-1) hash of the local IP address using an XOR operation. The data is then sent to the C2 server. ^[115]
S9020	LODEINFO	LODEINFO has used control flow flattening to obfuscate code. ^[116]
S0447	Lokibot	Lokibot has obfuscated strings with base64 encoding. ^[117]
S1213	Lumma Stealer	Lumma Stealer has used SmartAssembly to obfuscate .NET payloads. ^[118]
S0167	Matryoshka	Matryoshka obfuscates API function names using a substitute cipher combined with Base64 encoding. ^[119]
S0449	Maze	Maze has decrypted strings and other important information during the encryption process. Maze also calls certain functions dynamically to hinder analysis. ^[120]
S0500	MCMD	MCMD can Base64 encode output strings prior to sending to C2. ^[121]
S0051	MiniDuke	MiniDuke can use control flow flattening to obscure code. ^[80]
G1036	Moonstone Sleet	Moonstone Sleet delivers encrypted payloads in pieces that are then combined together to form a new portable executable (PE) file during installation. ^[122]
G0129	Mustang Panda	Mustang Panda has delivered initial payloads hidden using archives and encoding measures. ^{[123][124][125]} ^{[126][127][128][129][130][31]} ^{[131][132][133]} Mustang Panda has also utilized opaque predicates in payloads to hinder analysis. ^[134]
S0226	NanoCore	NanoCore 's plugins were obfuscated with Ezfuscator.NET 2.2. ^[135]

ID	Name	Description
S0353	NOKKI	NOKKI uses Base64 encoding for strings. ^[138]
S9025	NOOPLDR	NOOPLDR can use control flow flattening to help hide malicious code. ^{[139][140]}
S0138	OLDBAIT	OLDBAIT obfuscates internal strings and unpacks them at startup. ^[63]
S0264	OopsIE	OopsIE uses the Confuser protector to obfuscate an embedded .Net Framework assembly used for C2. OopsIE also encodes collected data in hexadecimal format before writing to files on disk and obfuscates strings. ^{[141][142]}
S0229	Orz	Some Orz strings are base64 encoded, such as the embedded DLL known as MockDll. ^[143]
S0594	Out1	Out1 has the ability to encode data. ^[144]
S0598	P.A.S. Webshell	P.A.S. Webshell can use encryption and base64 encoding to hide strings and to enforce access control once deployed. ^[145]
S0517	Pillowmint	Pillowmint has obfuscated the AES key used for encryption. ^[146]
S0124	Pisloader	Pisloader obfuscates files by splitting strings into smaller sub-strings and including "garbage" strings that are never used. The malware also uses return-oriented programming (ROP) technique and single-byte XOR to obfuscate data. ^[147]
S0013	PlugX	PlugX can use API hashing and modify the names of strings to evade detection. ^{[51][130]}
S0428	PoetRAT	PoetRAT has used a custom encryption scheme for communication between scripts. ^[148]
S0012	PoisonIvy	PoisonIvy hides any strings related to its own indicators of compromise. ^[149]
S0518	PolyglotDuke	PolyglotDuke can custom encrypt strings. ^[80]
S0150	POSHSPY	POSHSPY appends a file signature header (randomly selected from six file types) to encrypted data prior to upload or download. ^[150]
S0393	PowerStallion	PowerStallion uses a XOR cipher to encrypt command output written to its OneDrive C2 server. ^[151]
S1228	PUBLOAD	PUBLOAD has obfuscated DLL names using the xor13AddHash32 algorithm. ^[124]
S0196	PUNCHBUGGY	PUNCHBUGGY has hashed most its code's functions and encrypted payloads with base64 and XOR. ^[152]
S0197	PUNCHTRACK	PUNCHTRACK is loaded and executed by a highly obfuscated launcher. ^[153]
S0650	QakBot	QakBot has hidden code within Excel spreadsheets by turning the font color to white and splitting it across multiple cells. ^[154]
S0458	Ramsay	Ramsay has base64-encoded its portable executable and hidden itself under a JPG header. Ramsay can also embed information within document footers. ^[155]
S1130	Raspberry Robin	Raspberry Robin uses mixed-case letters for filenames and commands to evade detection. ^[156]
G1039	RedCurl	RedCurl has used malware with string encryption. ^[157] RedCurl has also encrypted data and has encoded PowerShell commands using Base64. ^{[158][159]} RedCurl has used <code>PyArmor</code> to obfuscate code execution of LaZagne . ^[158] Additionally, RedCurl has obfuscated downloaded files by renaming them as commonly used tools and has used <code>echo</code> , instead of file names themselves, to execute files. ^[160]

ID	Name	Description
S0240	ROKRAT	ROKRAT can encrypt data prior to exfiltration by using an RSA public key. ^{[25][164]}
S0148	RTM	RTM strings, network data, configuration, and modules are encrypted with a modified RC4 algorithm. ^{[165][166]}
S9037	RustyWater	RustyWater has an obfuscated function (i.e. love_me__()) that dynamically reconstructs the string WScript.Shell using hard-coded ASCII values and the Chr() function. ^[167]
S0446	Ryuk	Ryuk can use anti-disassembly and code transformation obfuscation techniques. ^[62]
S1018	Saint Bot	Saint Bot has been obfuscated to help avoid detection. ^[168]
S1099	Samurai	Samurai can encrypt the names of requested APIs. ^[169]
G0034	Sandworm Team	Sandworm Team has used Base64 encoding within malware variants. ^[170]
S1085	Sardonic	Sardonic can use certain ConfuserEx features for obfuscation and can be encoded in a base64 string. ^[171]
S0461	SDBbot	SDBbot has the ability to XOR the strings for its installer component with a hardcoded 128 byte key. ^[172]
S0596	ShadowPad	ShadowPad has encrypted its payload, a virtual file system, and various files. ^{[173][74]}
S9008	Shai-Hulud	Shai-Hulud has utilized double-base64 encoding to store stolen secrets within the Github Action Logs within the victim account. ^{[174][175][176][177]} Shai-Hulud has also leveraged three layers of base64 encoding of exfiltrated data for anti-forensic purposes. ^[178]
S0140	Shamoon	Shamoon contains base64-encoded strings. ^[179]
S0445	ShimRatReporter	ShimRatReporter encrypted gathered information with a combination of shifting and XOR using a static key. ^[180]
S0063	SHOTPUT	SHOTPUT is obscured using XOR encoding and appended to a valid GIF file. ^{[181][182]}
S0623	Siloscape	Siloscape itself is obfuscated and uses obfuscated API calls. ^[183]
S0633	Sliver	Sliver obfuscates configuration and other static files using native Go libraries such as <code>garble</code> and <code>gobfuscate</code> to inhibit configuration analysis and static detection. ^[184]
S1104	SLOWPULSE	SLOWPULSE can hide malicious code in the padding regions between legitimate functions in the Pulse Secure <code>libdsplibs.so</code> file. ^[185]
S1035	Small Sieve	Small Sieve has the ability to use a custom hex byte swapping encoding scheme combined with an obfuscated Base64 function to protect program strings and Telegram credentials. ^[186]
S1086	Snip3	Snip3 has the ability to obfuscate strings using XOR encryption. ^[187]
S0627	SodaMaster	SodaMaster can use "stackstrings" for obfuscation. ^[77]
S0615	SombRAT	SombRAT can encrypt strings with XOR-based routines and use a custom AES storage format for plugins, configuration, C2 domains, and harvested data. ^{[188][189][190]}
S0516	SoreFang	SoreFang has the ability to encode and RC6 encrypt data sent to C2. ^[191]

ID	Name	Description
S0562	SUNSPOT	SUNSPOT encrypted log entries it collected with the stream cipher RC4 using a hard-coded key. It also uses AES128-CBC encrypted blobs for SUNBURST source code and data extracted from the SolarWinds Orion <MsBuild.exe process. ^[196]
S1064	SVCRReady	SVCRReady can encrypt victim data with an RC4 cipher. ^[197]
S0242	SynAck	SynAck payloads are obfuscated prior to compilation to inhibit analysis and/or reverse engineering. ^[198] ^[199]
S0467	TajMahal	TajMahal has used an encrypted Virtual File System to store plugins. ^[200]
S0560	TEARDROP	TEARDROP created and read from a file with a fake JPG header, and its payload was encrypted with a simple rotating XOR cipher. ^[195] ^[201] ^[202]
S0266	TrickBot	TrickBot uses non-descriptive names to hide functionality. ^[203]
S0094	Trojan.Karagany	Trojan.Karagany can base64 encode and AES-128-CBC encrypt data prior to transmission. ^[204]
S0647	Turian	Turian can use VMProtect for obfuscation. ^[29]
S0476	Valak	Valak has the ability to base64 encode and XOR encrypt strings. ^[205] ^[206] ^[207]
G0112	Windshift	Windshift has used string encoding with floating point calculations. ^[208]
S0117	XTunnel	A version of XTunnel introduced in July 2015 obfuscated the binary using opaque predicates and other techniques in a likely attempt to obfuscate it and bypass security products. ^[209]

Mitigations

ID	Mitigation	Description
M1049	Antivirus/Antimalware	Anti-virus can be used to automatically detect and quarantine suspicious files. Consider utilizing the Antimalware Scan Interface (AMSI) on Windows 10+ to analyze commands after being processed/interpreted. ^[210]
M1047	Audit	Consider periodic review of common fileless storage locations (such as the Registry or WMI repository) to potentially identify abnormal and malicious data.
M1040	Behavior Prevention on Endpoint	On Windows 10+, enable Attack Surface Reduction (ASR) rules to prevent execution of potentially obfuscated payloads. ^[211]
M1017	User Training	Ensure that a finite amount of ingress points to a software deployment system exist with restricted access for those required to allow and enable newly deployed software.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0378	Behavioral Detection of Obfuscated Files or Information	AN1064	Correlates script execution or suspicious parent processes with creation or modification of encoded, compressed, or encrypted file formats (e.g., .zip, .7z, .enc) and abnormal command-line syntax or PowerShell obfuscation.
		AN1065	Detects use of gzip, base64, tar, or openssl in scripts or commands that encode/encrypt files after file staging or system enumeration.
		AN1066	Monitors use of archive or encryption tools (zip, openssl) tied to user-scripted activity or binaries writing encoded payloads under /Users or /Volumes.
		AN1067	Identifies transfer of base64, uuencoded, or high-entropy files over HTTP, FTP, or custom protocols in lateral movement or exfiltration streams.
		AN1068	Detects encoded PowerCLI or Base64-encoded payloads staged via datastore uploads or shell access (e.g., ESXi Shell or backdoored VIBs).

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